

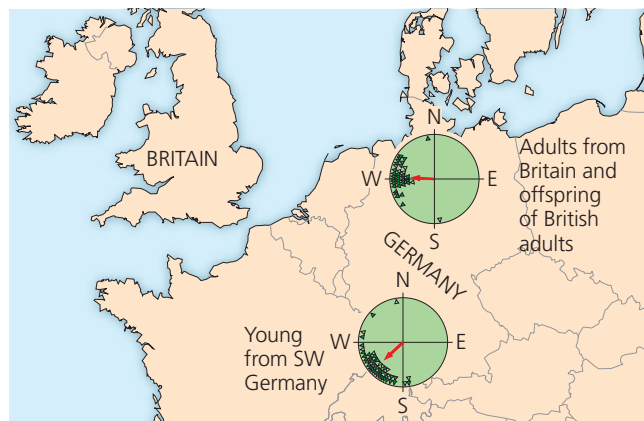
▼ Figure 51.24 Inquiry

Are differences in migratory orientation within a species genetically determined?

Experiment Birds known as blackcaps that live in Germany winter elsewhere. Most migrate to Spain and Africa, but a few fly to Britain, where they find food left out by city dwellers. German scientist Peter Berthold and colleagues wondered if this change had a genetic basis. To test this hypothesis, they captured blackcaps wintering in Britain and bred them in Germany in an outdoor cage. They also collected young blackcaps from nests in Germany and raised them in cages. In the autumn, the blackcaps captured in Britain and the birds raised in cages were placed in large, glass-covered funnel cages. When the funnels were lined with carbon-coated paper and placed outside at night, the birds moved around, making marks on the paper that indicated the direction in which they were trying to “migrate.”



Results The wintering adult birds captured in Britain and their laboratory-raised offspring both attempted to migrate to the west. In contrast, the young birds collected from nests in southern Germany attempted to migrate to the southwest.



Conclusion The young of the British blackcaps and the young birds from Germany (the control group) were raised under similar conditions but showed very different migratory orientations, indicating that their migratory orientation has a genetic basis.

Data from P. Berthold et al., Rapid microevolution of migratory behavior in a wild bird species, *Nature* 360:668–690 (1992).

WHAT IF? Suppose the birds had not shown a difference in orientation in these experiments. Could you conclude that the behavior was not genetically based? Explain.

▼ **Figure 51.25** Naked mole rats, a species of colonial mammal that exhibits altruistic behavior. Pictured here is a queen nursing offspring while surrounded by other members of the colony.



Altruism is also observed in naked mole rats (*Heterocephalus glaber*), highly social rodents that live in underground chambers and tunnels in southern and northeastern Africa. The naked mole rat, which is almost hairless and nearly blind, lives in colonies of 20 to 300 individuals (Figure 51.25). Each colony has only one reproducing female, the queen, who mates with one to three males, called kings. The rest of the colony consists of nonreproductive females and males who at times sacrifice themselves to protect the queen or kings from snakes or other predators that invade the colony.

Inclusive Fitness

With these examples from ground squirrels, honeybees, and mole rats in mind, let's return to the question of how altruistic behavior arises during evolution. The easiest case to consider is that of parents sacrificing for their offspring. When parents sacrifice their own well-being to produce and aid offspring, this act actually increases the fitness of the parents because it maximizes their genetic representation in the population. By this logic, altruistic behavior can be maintained by evolution even though it does not enhance the survival and reproductive success of the self-sacrificing individuals.

What about circumstances when individuals help others who are not their offspring? By considering a broader group of relatives than just parents and offspring, biologist William Hamilton found an answer. He began by proposing that an animal could increase its genetic representation in the next generation by helping close relatives other than its own offspring. Like parents and offspring, full siblings have half their genes in common. Therefore, selection might also favor helping siblings or helping one's parents produce more siblings. This thinking led Hamilton to the idea of **inclusive fitness**, the total effect an individual has on proliferating its genes by producing its own offspring and by providing aid that enables other close relatives to produce offspring.